Orchard Rain and Snowmelt Harvesting Plan

Prairie Plants · Ideas from Brad Lancaster's *Rainwater Harvesting for Drylands and Beyond* (Vol. 1, 3rd ed., and Vol. 2, 2nd ed., both 2019), adapted to a snowy site with dry summers

Companion page; it does not change the planting or pollination plans. Tree positions use the current canonical planting records in orchard_pollination_grid.html. Slope from USGS 3DEP lidar, the same data as the whole-site maps: the homestead is the high ground, the land falls north and west to the low northwest corner by the road, and the planted block sits at the south end of the 600 ft strip, with the R1 trunks on its south line (October 24, 2026 tape). Climate from PRISM 1991–2020 normals and ERA5 reanalysis (see Sources and Assumptions).

PRECIPITATION	FALLS NOV–MAR	FALLS JUL–SEP	MAY–SEP DEMAND	SOIL
22 in/yr	**64%**	**1.2 in**	**30.6 in**	**Simonton loam**
PRISM normal; NRCS lists 12–16 in for this soil	14.1 in; about half of the year comes as snow	for the whole summer	reference ET_0, vs 4.2 in of rain	loam over clay loam; subsoil passes about 0.4 in/hr

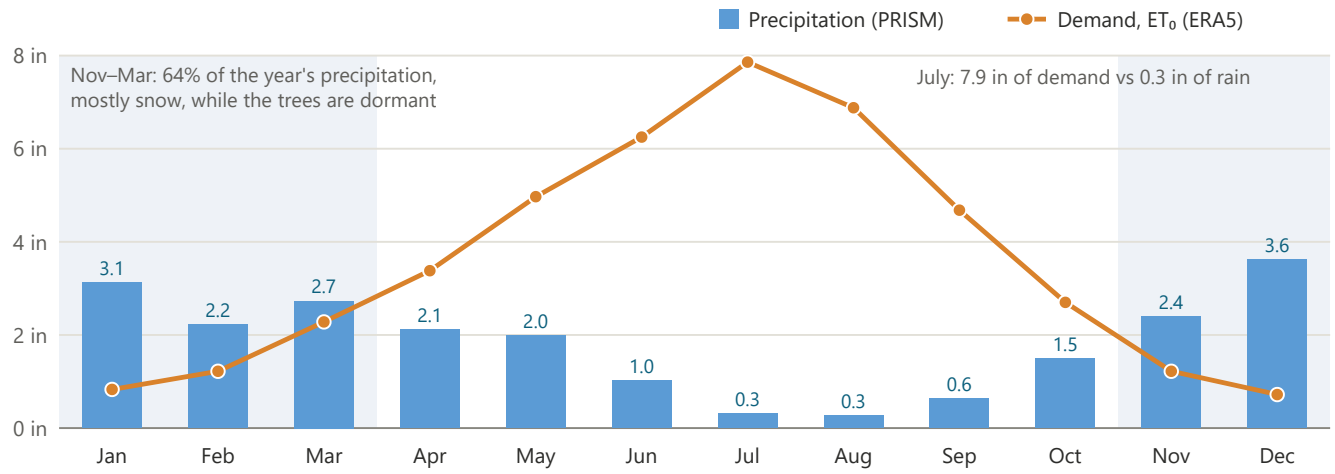
LOW POINT	WATER FLOWS TOWARD	TYPICAL GRADE	EACH TREE'S 18 × 20 FT	PLANTED BLOCK
NW corner	**N 8–66° W**	**3.5%**	**≈ 4,900 gal**	**≈ 970,000 gal**
just under 1445 m, by the road, 320 ft north of R15	across the block; median N 32° W	2.3%–4.5% across the block (lidar)	of rain and snow water a year, gross	a year on 1.6 acres, gross

The Plan in Brief

PRISM puts Prairie at about **22 in** of precipitation a year, about double the ~11 in often quoted for Elmore County, which reflects the low desert around Mountain Home. NRCS lists 12–16 in for your soil's setting, so your lot may get less; a rain gauge will settle it. Either way it's more than Lancaster gets in Tucson. The catch is timing: about two-thirds falls from November through March, and about half of the year's total arrives as snow while the trees are dormant. July through September brings only about an inch, just when a mature orchard's demand peaks. The plan:

- **Start at the top.** The homestead roofs and gravel pad sit at the high point just south of the orchard. Catch their water and spread it into the top rows.
- **Catch it where it lands.** A basin at every tree soaks snowmelt and spring rain deep, filling the soil "tank" before summer.
- **Pass the overflow downhill.** Each basin spills to the next tree, and the chain ends in a sponge basin at the low northwest corner by the road.
- **Let the drip fill the July–September gap.** Harvested water won't replace the drip, but it can delay the first run, shorten the season, and soften dry spells.

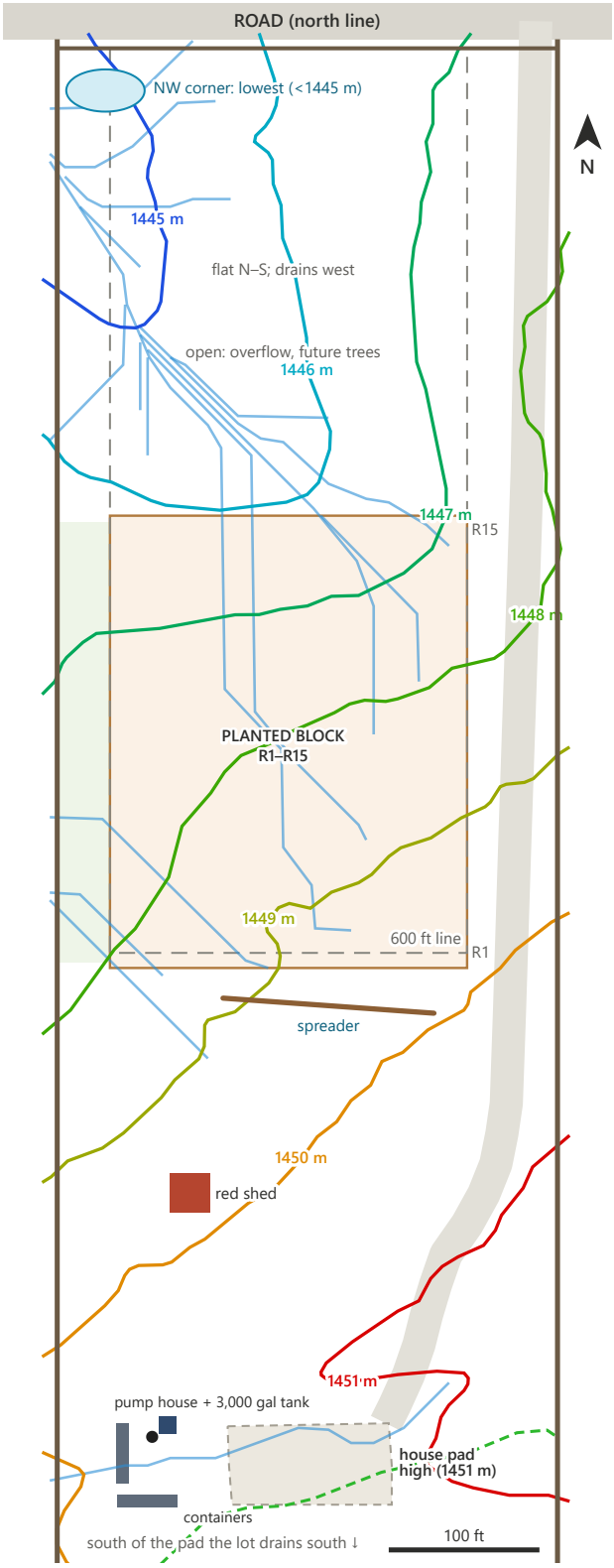
Water In vs. Water Demand, by Month



Bars: PRISM 1991–2020 normal precipitation (inches of water) at Prairie. Line: FAO-56 reference evapotranspiration (ET_0) from ERA5, 1995–2024, the water a well-watered grass surface would use; mature trees with full canopies use a similar amount per area of canopy, young trees far less. In the shaded months the mean temperature is below about 36 °F, so most of that precipitation arrives as snow.

The Land: Lidar Contours and Flow Paths

Whole field, north up like the site maps



- The lidar puts the high ground at the homestead and the east fence beside it (about 1,451.4 m) and the low point at the west fence near the northwest corner (about 1,444.5 m): about 22 ft of fall. Across the north end the west tilt is about as strong as the north fall.
- The north part of the strip, the 320 ft beyond R15, is nearly flat north–south and drains west at about 4.5% toward the west fence and the NW corner. That is where the overflow ends up.
- The block falls northwest in R1–R5 (about N 48° W) and north-northwest in R9–R12 (about N 14° W), at a typical 3.5% (2.3%–4.5%). The planted trees span 8.6 ft of elevation, and R1's east end is 5.4 ft above its west end, so every row drains west.
- The rows are offset by 9 ft, so each tree's northwest diagonal neighbor sits 24° west of north, close to the median fall. On the site plan each basin's overflow goes to whichever neighboring tree is downhill, usually that diagonal; long swales along the rows would just carry water to their west ends.
- The stone fruit (R9–R12) sits on the steepest band (typical 3.9%), where water and cold air keep moving. The lowest planted tree, R12-C1 (apricot), is still about 8 ft above the corner, which is the frost pocket.
- The lidar flow paths (blue) show where runoff concentrates: the main one enters the block near the middle of R1 and runs north-northwest through it, then on to the west fence just south of the NW corner, with shorter paths joining it. Those paths are where basins catch the most.

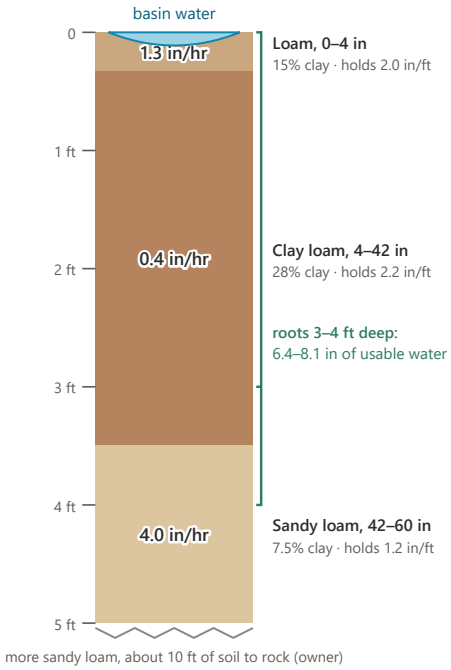
1 m contours (lidar) · Flow path (lidar) · Divide (runoff splits)
Planted block · 600 ft orchard strip
Pump house and irrigation tank · NW sponge basin

From USGS 3DEP lidar fitted to the fences (331 × 1,292 ft), shared with the whole-site terrain and water maps. Contours are 1 m (3.3 ft) apart, labeled in meters. Blue lines are where runoff from a tenth of an acre or more gathers; south of the green split line the lot drains south, away from the orchard. The shed, containers, pump house, 3,000 gal irrigation tank, and house pad are placed from the site master grid (drone photos, October 2026).

The Soil: Simonton Loam and How Water Moves Through It

USDA NRCS Web Soil Survey report for the property: map unit 149, Simonton loam, 2 to 12 percent slopes (Elmore County Area, Idaho, survey version 14, August 31, 2026), with layer data from NRCS Soil Data Access. Owner: no rock on the property and about 10 ft of soil before rock.

Typical profile, to scale



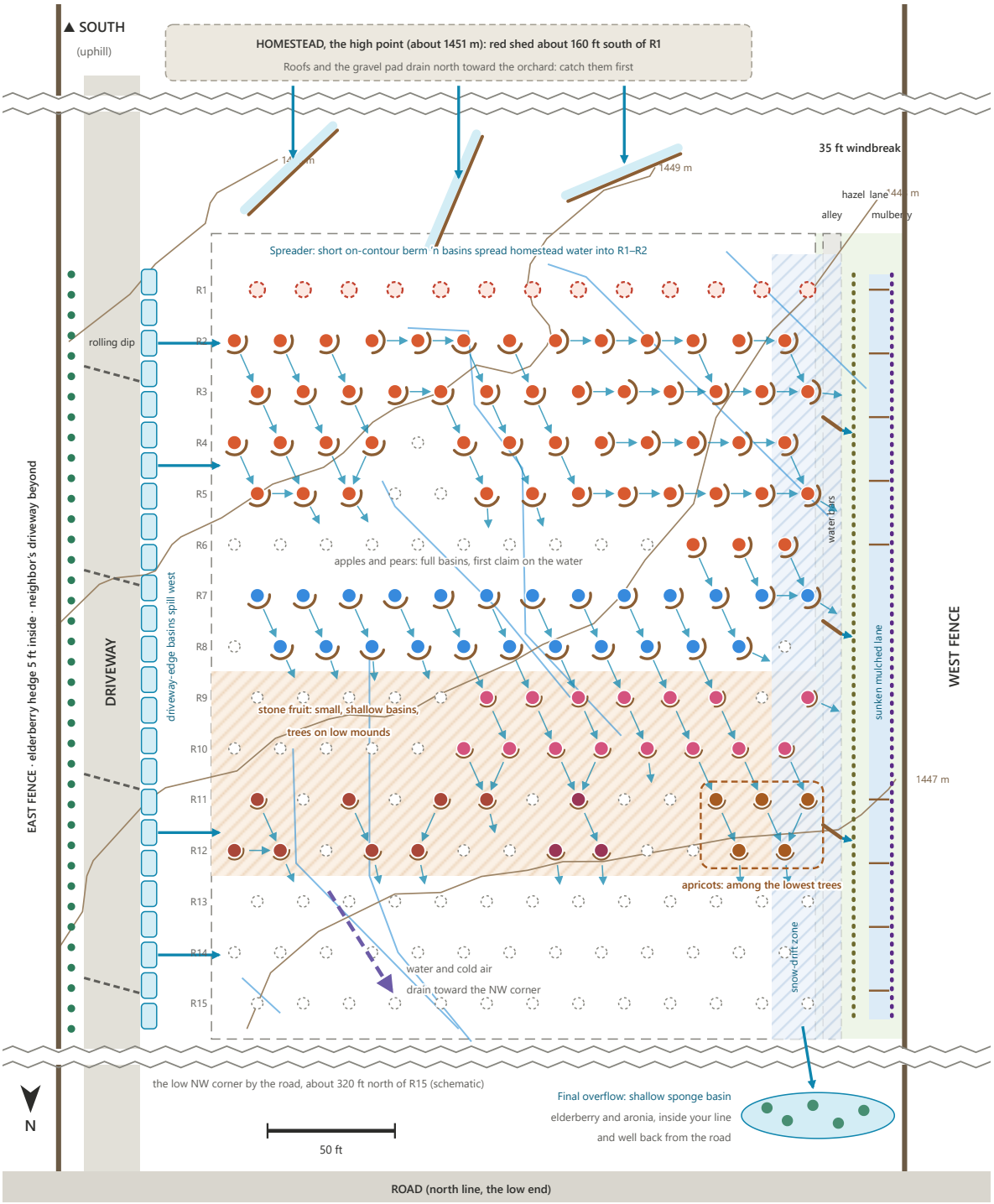
- Fast top, slow middle. The top 4 in of loam soaks up water at about 1.3 in/hr, but the clay loam below passes only about 0.4 in/hr (0.2–0.6). A basin wets the topsoil within minutes, then drains at the subsoil's pace: 4 in of water is gone in about 10 hours (7–20). That's why basins here hold 3–4 in, not more.
- Keep basin floors in the loam. NRCS's typical profile has only about 4 in of loam over the clay loam (tilling mixes the top several inches; a test hole shows where the stickier layer starts). Bare clay loam crusts and seals, so dig no deeper than about 2 in, put any topsoil back, and get the depth from the berm and a low planting mound.
- A big soil tank. The top 3–4 ft holds about 6.4–8.1 in of water roots can use (9.3 in in the top 5 ft), about 1,400–1,800 gal under each tree's 18 × 20 ft. Filling it from dry takes that much water actually soaking in, a big share of what falls November through June once blowing snow, sublimation, and runoff take their cut. Catching runoff is how you fill it.
- Hydrologic group C. Once it's wet, or frozen in March, this soil sheds a lot of water. That runoff is exactly what the spreader, the crescents, and the driveway basins are for.
- Deep and rock-free. Below about 42 in, sandy loam drains freely (about 4 in/hr); NRCS finds no hardpan or water table within 80 in, and you have about 10 ft of soil to rock. Roots can go deep and nothing waterlogs them from below, but water that sinks past the roots is lost to the trees, so run the drip in moderate pulses rather than long soaks.
- Spring is the wet-feet window. Snowmelt can sit on the clay loam for a while before it soaks in, the cold, wet conditions crown rot and canker like. Keep stone-fruit basins at 3 in or less and those trees on mounds; by summer the profile drains well.
- Friendly chemistry: pH about 6.2, no lime, no salt. Organic matter is low (about 1.5% on top, less below), so chips and cover crops build it and keep the surface open. Bare, worked loam here is also moderately wind-erodible, one more reason to keep it covered.
- Rain check: NRCS lists 12–16 in a year for this soil's setting, against PRISM's 22 in at Prairie. If your lot is nearer the NRCS figure, every gallon on the budget page shrinks by a quarter to almost half, and harvesting matters more, not less. A rain gauge settles it.

Layer	Depth	Texture	Clay	Soaks in, in/hr	Holds, in per ft	What it means for water
A	0–4 in	Loam	15%	1.3 (0.6–2.0)	2.0	The sponge layer: most of the organic matter (about 1.5%), and it takes water fast. Keep it in basin floors and keep it covered.
Bt	4–42 in	Clay loam	28%	0.4 (0.2–0.6)	2.2	The bottleneck: it sets how fast basins drain, and it holds most of the root zone's water.
C	42–60+ in	Sandy loam	7.5%	4.0 (2.0–6.0)	1.2	Drains freely: no hardpan or water table, and about 10 ft of soil to rock (owner).

"Soaks in" is NRCS's saturated hydraulic conductivity (Ksat), typical value with the low–high range; "holds" is available water capacity, the water roots can use. Hydrologic soil group C; well drained; no flooding or ponding; no water table or restrictive layer within 80 in.

Water-Harvesting Site Plan — To Scale, South Up, West Fence on the Right

Same orientation as the planting map (the view from the north end looking south), so the homestead, the high point, is at the top and the road, the low end, is at the bottom. Distances are from the west fence and the R1 row line. Hover over a tree for its position.



Crescent berm (full basin) · Small, shallow basin (stone fruit) · Overflow to the next tree downhill · 1 m contour (lidar, label in meters)
Flow path (lidar) · Spreader berm 'n basin (homestead water) · Driveway-edge or sponge basin · Water bar or check berm
Rolling dip across the driveway · Sunken, mulched walking lane · Likely snow-drift zone · Stone-fruit rows
Fall line: water and cold air · Apple · Pear · Cherry · Peach · Apricot · Plum
Reserved (grafted apples) · Open position

Contours (1 m apart, about 3.3 ft) and blue flow paths come from USGS lidar, with R1 on the south end of the 600 ft strip, as taped on October 24, 2026. Each crescent faces the local downhill direction from that data, and each overflow arrow runs to the neighboring tree that is downhill; a short arrow means no planted neighbor is downhill. Crescents are drawn at young-tree size, about 4–5 ft out. The homestead and the road are schematic beyond the breaks. The lidar surface is smoothed to about 5 ft and was flown before recent grading, so confirm the fall at each tree with a level before building.

Techniques by Zone, Top of the Slope to the Bottom

Lancaster's second principle: start at the top of the watershed and work down. Book references are chapters in Volume 1 (3rd ed.) or Volume 2 (2nd ed.).

Zone	What to build	Size and specs	Book	When
1. Homestead (the high point)	Start here. Gutter the red shed, the two containers, the pump house, and any future house into tanks for early-summer watering, or straight into the spreader below. Let winter snow slide off roofs into basins, not onto paths. Send the gravel pad's runoff north in a gently sloped diversion swale to the spreader.	Each 1,000 sq ft of roof sheds about 620 gal per inch, about 4,900 gal from March through June. A tank at the homestead sits above the orchard rows and gives about 0.43 psi per foot of drop: enough for low-pressure gravity drip, not for your 15 psi drip tape without a pump. Use the calculator on the budget page for your own areas.	Vol. 1 Ch. 3 (Cistern Systems); Vol. 2 Ch. 8 (Hardscape Runoff), Ch. 9 (Diversion Swales)	Spring–summer 2027
2. Spreader above R1	Catch the homestead water, and any sheet flow from upslope, in short, staggered on-contour berm 'n basins across the top of the block, spilling into the R1–R2 basins. The apples there are the best-drained trees and the best able to use it. Contours here cross the rows at a slant (17–56°), so a single long contour line would cut through rows: keep each segment short and level.	Segments 30–60 ft long, 3–4 in of water depth, level crests, spillways at the ends lined with sod (no rock needed)	Vol. 2 Ch. 3 (Berm 'n Basins), Ch. 9	With the first apples
3. Every orchard tree	A crescent berm 'n basin on the downhill side (northwest to north-northwest here), with its spillway aimed at the next tree downhill, usually the northwest diagonal. Plant on the basin's upper edge, never in its bottom. Put the drip emitters inside the basin.	Start 3–4 ft from the trunk; widen toward the drip line as the tree grows. Tree on a low mound at the upper edge; level floor dug no deeper than about 2 in, so it stays above the slower clay-loam subsoil. Spillway 3–4 in above the floor (3 in or less for stone fruit), lined with sod; berm crest level and 2–3 in above it; low and rounded so you can mow over it.	Vol. 2 Ch. 3 (Berm 'n Basins)	At planting (now)
4. Orchard floor	Living sponge: wood chips in the basins (chip the windbreak and orchard prunings), perennial grass-and-clover cover in the alleys, clippings mowed or blown onto the tree rows. Mulched soil freezes less deeply, so snowmelt can soak in instead of sheeting off frozen ground. Stay off soft rows during snowmelt; ruts become ditches running west.	Chips 2–3 in deep in basins (about 2 in for stone fruit), surface at least 1 in below the spillway, kept 6–12 in back from trunks; 3–4 in elsewhere. Hardware-cloth trunk guards: mulch plus snow cover hides voles.	Vol. 2 Ch. 7 (Mulching)	At planting; top up yearly
5. Driveway (east side)	It runs downhill along the orchard, and the land tilts west, so its runoff, plus some from the higher neighbor's side, heads into the headland. Catch it in a chain of shallow, mowable basins along the driveway's west edge, stepped down the slope by low check berms, each spilling west to the east-end (C13) trees. If the drive is gravel, broad rolling dips across it kick water into the basins. Push salt-free plowed snow into the basins, not onto young plants.	Basins about 6 ft wide with 3–4 in of water depth; the drive falls about 3% along the block, so a check berm about every 12–15 ft keeps each cell level. Each 100 ft of the 22 ft drive sheds up to about 1,370 gal per inch of rain. Never route salted snow or oily runoff to plants.	Vol. 2 Ch. 8; Ch. 5 (Infiltration Trenches)	Spring–summer 2027
6. Elderberry hedge	It sits upslope of most of the driveway's runoff, so it lives on its own rain and snow. If the drive is crowned and its east half drains east, small crescents on each bush's downhill side catch that water.	Crescents 2–3 ft out, 3–4 in of water depth	Vol. 2 Ch. 3	As the cuttings root
7. West windbreak (snow fence)	Snow-drift harvesting: west winds will drop drifts just east of the hazelnut row, over the tractor alley and the C1 berry guilds. Let them melt in place. Brush heavy snow off young limbs, and expect the alley to stay blocked until the drift melts.	Drifts grow with the windbreak. A dense, bushy bottom piles snow close; a more open row spreads it farther and thinner.	Vol. 1 App. 8; Vol. 2 Ch. 3 ("Harvesting Windswept Snow")	Every winter
8. Tractor alley	Low, broad, drivable diagonal humps (water bars) so wheel ruts can't turn into a gully running north. Each one turns water west into the hazelnut row.	Every 75–100 ft; 4–6 in high; angled about 30° across the alley	Vol. 2 Ch. 9 (Diversion Swales)	When ruts appear
9. Walking lane	Make the mulberry–hazelnut lane a sunken wood-chip path between the two planted rows, so it soaks up overflow for both. Low cross-berms keep water from running north along it.	Cross-berms 4–6 in high every 20–25 ft (the west side falls about 2%)	Vol. 1 Ch. 4, pattern 6 (Raised Paths, Sunken Basins)	When the lane is laid out

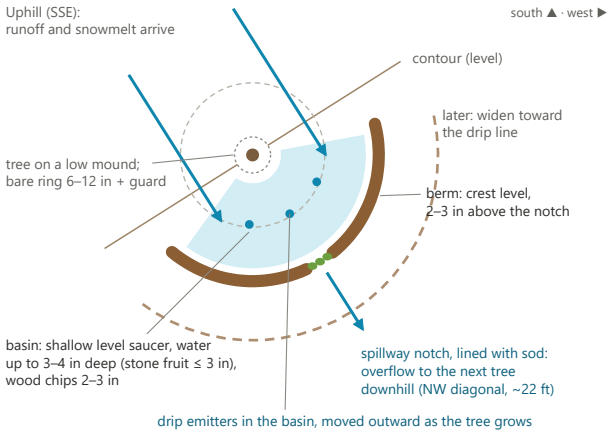
Zone	What to build	Size and specs	Book	When
10. Northwest corner (by the road)	The final overflow "sponge": a broad, shallow infiltration basin planted with wet-tolerant, later-blooming shrubs such as elderberry (from your hardwood cuttings) and aronia. Keep it inside your line and back from the road so overflow never cuts the road edge, and keep it in the corner rather than a solid hedge across the north end, so cold air can still drain away.	6 in deep or less and wide; size it after watching one snowmelt; give it a spillway too.	Vol. 2 Ch. 2 (Infiltration Basins)	After a season of watching

Lancaster's Eight Principles, Applied Here

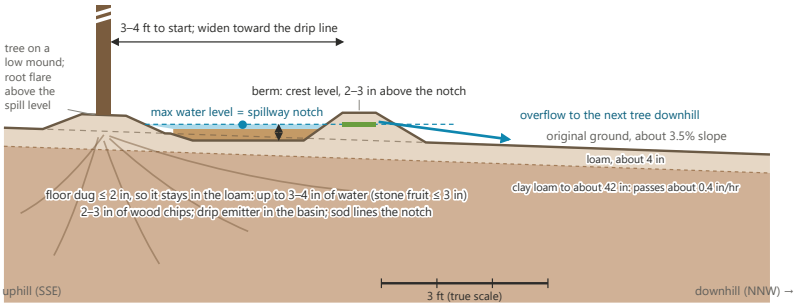
#	Principle (Lancaster's wording)	On your site
1	Begin with long and thoughtful observation.	Walk the block during snowmelt and after summer storms. Flag where water runs and ponds and where snow and frost linger, and use the log on the last page.
2	Start at the top (highpoint) of your watershed and work your way down.	Your high point is the homestead pad: its roofs and gravel come first, then the driveway and the top rows. The northwest corner by the road comes last.
3	Start small and simple.	One crescent per tree: at planting for new trees, and as soon as you can for the trees already in the ground. Add the spreader, driveway basins, lane, and corner basin after a season of watching.
4	Slow, spread, and infiltrate the flow of water.	Level basin floors and level berm crests; many small basins instead of one ditch; mulch everywhere water lands.
5	Always plan an overflow route, and manage that overflow as a resource.	Every basin spills to the next tree downhill, and the chain ends in the northwest sponge basin, not the road.
6	Maximize living and organic groundcover.	Wood chips in the basins, perennial cover in the alleys, clippings blown onto the rows.
7	Maximize beneficial relationships and efficiency by "stacking functions."	The windbreak is also a snow fence, a nut crop, and a mulch source; the berry guilds sit in the drift zone; elderberry cuttings plant the sponge basin; roof tanks double as gravity-fed drip.
8	Continually reassess your system: the "feedback loop."	Each fall, widen basins toward the drip line and fix spots where water broke through or stood too long.

Crescent Berm 'n Basin at Each Tree

Plan view, oriented like the site plan (south up, west on the right)



Section down the fall line (uphill on the left), drawn to true scale



Build it as you plant

1. Plant on a low mound, 3–4 in high and wider than the root ball, so the root flare ends up 1–2 in above the future spillway level after the soil settles. (The fall 2026 trees are already planted, so start at step 2 for them.)
2. Find the downhill direction at the tree with a level. Expect northwest in R1–R5 and north-northwest farther north (the site plan shows each tree), but local bumps rule.
3. Mark an arc on the downhill side, 3–4 ft from the trunk, with its ends running along the contour and curving slightly uphill.
4. Shape a shallow, level-floored saucer between the mound and the arc, digging no deeper than about 2 in so the floor stays in the loam above the stickier clay-loam subsoil. Get the rest of the depth from the berm: pile soil on the arc (scrape extra from the alley side, never from the basin floor), tamp it, and level its crest.
5. Cut a spillway notch in the middle of the berm so water can stand 3–4 in deep at most (3 in or less for stone fruit), with the crest 2–3 in higher. You don't need rock: line the notch with a strip of sod from the alley, or a paver or board set flush, so overflow can't cut it. Aim it at the next tree downhill, usually the northwest diagonal about 22 ft away (the site plan shows which).
6. Mulch the basin with 2–3 in of wood chips (about 2 in for stone fruit), keeping the chip surface at least 1 in below the spillway and a 6–12 in bare ring around the trunk, and add a hardware-cloth guard.
7. Set the drip emitters inside the basin under the outer canopy, and move them outward as the tree grows.
8. Test with a hose or the first storm. Water should spread along the berm, not pour out one end, and soak in within about a day.
9. Each fall, extend the arc outward toward the drip line.

Find the contour

An A-frame level, a water level (clear tubing filled with water), or a laser level all work; Volume 2, Appendix 1 covers all three. With an A-frame, mark where the plumb line crosses the crossbar, flip the frame end for end on the same two spots, and mark again; halfway between the marks is level. Then walk the frame across the slope, swinging one leg until the line hits the level mark, and flag each foot. The flags trace a contour. On your ground, expect contours to run east-northeast to northeast, crossing the tree rows at about 8–66°.

Test how fast your soil drinks

Dig a hole about 12 in deep and 8 in wide near the south end, the middle, and the northwest part of the block; its sides also show where the stickier clay loam starts. Fill it, let it drain, refill it, and measure how far the water drops in one hour. NRCS rates the clay-loam subsoil at about 0.4 in per hour (0.2–0.6), so 4 in of water should be gone in about 10 hours (7–20). For a basin to empty within about a day, keep its water depth no more than 24 times your hourly drop. Where water still stands after a day, send extra water only to wet-tolerant plants.

Who Gets the Water

Counts and positions come from the current planting records, including the planted M-111 rootstock; vacant spots and outside-orchard trees are excluded. Wet-soil tolerance follows UC IPM's fruit site-selection guidance; canker notes follow the inventory flags shown on the pollination grid.

Fruit	Where · trees	Wet-soil tolerance	Basin	Send overflow?	Notes
Apples	R2–R6 · 52 (+13 reserved in R1)	Moderate	Full crescent; spillway 3–4 in above the floor	Yes: first claim	The highest, best-drained trees, first in line for the homestead water. The 13 reserved R1 spots get basins when the grafted apples go in.
Pears	R7–R8 · 24	Best of your fruit trees (UC IPM)	Full crescent; spillway 3–4 in above the floor	Yes: can take the most	A natural catch for overflow coming down from the apples.
Cherries	R9–R10 · 15	Low: can't take saturated soil (UC IPM)	Small; spillway 3 in or less; tree on a low mound	No: let run-on pass by	Bacterial canker is flagged on several; UC IPM says it is worse in low spots and after spring freezes.
Peaches	R11–R12, C8–C13 · 8	Low: can't take saturated soil (UC IPM)	Small; spillway 3 in or less; tree on a low mound	No	Canker flagged on most; early bloom.
Apricots	R11–R12, C1–C3 · 5	Low (stone fruit)	Smallest; if the perc test is slow, add a low uphill berm that steers run-on around the tree	No	The lowest planted trees and the earliest bloom; canker flagged on the apricots.
Plums	R11–R12, C5–C6 · 3	Low to moderate	Small; spillway 3 in or less	Little	No canker flags in the inventory.
Berries, windbreak, hedge	C1 guilds, west strip, east hedge	Elderberry and aronia handle wet soil; currants, gooseberries, jostaberry, and honeyberry like moist but drained soil; hazelnut prefers drained soil; mulberry is adaptable	Their own crescents; the lane and the NW basin take overflow	Yes: the west edge and the NW corner are where overflow ends	The C1 guilds sit in the snow-drift zone.

Cold air follows the water. On frosty spring nights cold air drains the same way water does, toward the northwest corner by the road. With the block at the south end of the strip, that open corner, about 300 ft past R15, is the frost pocket, not the orchard. Within the block the stone fruit (R9–R12) is on the lower half but on its steepest band, so cold air keeps moving, and the apricots (R11–R12, C1–C3) and the plums at R12-C5 and R12-C6 are the lowest planted trees. Stone fruit blooms first, and UC IPM notes that bacterial canker, already flagged on many of these trees, is worse in low spots and after spring freezes; drought stress and irrigation that wets the trunk also set trees up for it. So these trees need water that soaks through, never water that stands, plus steady drip in summer. Don't plant a solid hedge across the block's north end, where it would dam cold air against them. NRCS lists only 60–100 frost-free days for this soil's setting, so that cold-air drainage matters.

Row-by-Row Build Tracker

Row	Trees in the plan (west to east)	Basin	Crescents shaped	Chips + guards	Drip in basins	Notes
R1	13 positions reserved for grafted apples	Full, when planted	☐	☐	☐	
R2	13 apples	Full: spillway 3–4 in	☐	☐	☐	
R3	13 apples	Full: spillway 3–4 in	☐	☐	☐	
R4	12 apples (1 open)	Full: spillway 3–4 in	☐	☐	☐	
R5	11 apples (2 open)	Full: spillway 3–4 in	☐	☐	☐	
R6	3 apples (10 open)	Full: spillway 3–4 in	☐	☐	☐	
R7	13 pears	Full: spillway 3–4 in	☐	☐	☐	
R8	11 pears (2 open)	Full: spillway 3–4 in	☐	☐	☐	
R9	7 cherries (6 open)	Small: spillway ≤ 3 in, low mounds	☐	☐	☐	
R10	8 cherries (5 open)	Small: spillway ≤ 3 in, low mounds	☐	☐	☐	
R11	3 apricots, 1 plum, 4 peaches (5 open)	Small: spillway ≤ 3 in, low mounds	☐	☐	☐	
R12	2 apricots, 2 plums, 4 peaches (5 open)	Small: spillway ≤ 3 in, low mounds	☐	☐	☐	
R13	Open for future trees	—				
R14	Open for future trees	—				
R15	Open for future trees	—				

Water Budget (Lancaster's Formula)

Gallons = area (sq ft) × inches of precipitation × 0.623 (Vol. 1 Ch. 2 and App. 3). These are gross volumes, before runoff, evaporation, and snow sublimation; the earthworks exist to keep more of it on your ground. They use PRISM's 22 in; if your gauge shows NRCS's 12–16 in, multiply by about 0.55–0.75.

Catchment	Area, sq ft	Gal per inch	Gal per average year	Nov–Mar	Apr–Jun	Jul–Sep
One tree's space (18 × 20 ft)	360	224	4,900	3,200	1,100	270
One full row of 13 trees	4,680	2,920	63,900	41,100	14,900	3,600
Planted block (237 × 300 ft)	71,100	44,300	970,000	620,000	230,000	54,000
Driveway, per 100 ft (22 ft wide)	2,200	1,370	30,100	19,300	7,000	1,700
Roof, per 1,000 sq ft	1,000	623	13,700	8,800	3,200	760

Your homestead catchments (type in the areas)

Measure each roof's footprint including the eaves, and the pad. The runoff factor is the rough share that runs off: about 0.9 for metal roofs and 0.5 for compacted gravel; edit to suit. March through June is the tank-filling season; the year total is gross.

Catchment	Area, sq ft	Runoff factor	Gal per inch	Gal Mar–Jun	Gal per year
Red shed roof	sq ft	0.9	—	—	—
Container roofs (two)	sq ft	0.9	—	—	—
Pump house roof	sq ft	0.9	—	—	—
House roof (future)	sq ft	0.9	—	—	—
Gravel house pad	sq ft	0.5	—	—	—
Other	sq ft	0.5	—	—	—
Total			**—**	**—**	**—**

Summer gap for one mature tree (rough)

On one tree's 18 × 20 ft space	Inches	Gallons
July rain (PRISM)	0.31	70
July reference demand, ET_0 (ERA5)	7.86	1,800
Mature tree in July, about 0.8 × ET_0	6.3	1,400 (about 45 a day)
May–September rain	4.2	950
Mature tree, May–September, about 0.8 × ET_0	24.5	5,500
Soil "tank": 3–4 ft of roots in Simonton loam, full	6.4–8.1	1,400–1,800

Summer rain covers only a sliver of a mature tree's use. A full soil tank can carry a mature tree about 2–3 weeks into peak summer before it feels stress (the first half of the tank), and the drip covers the rest. Young trees use far less but reach only the soil near the trunk, which is why their basins start small and grow outward.

Season-by-Season Calendar

When	What's happening	Water-harvesting work
Oct–Nov (now)	Planting; first snow (November averages 2.4 in of water)	Shape a crescent as each tree goes in; add chips and trunk guards; flag the fall line at a few trees; run a perc test in three spots before the ground freezes. If fall stays dry, give one deep drip watering before freeze-up. Set up a rain gauge; CoCoRaHS volunteers report daily, snow included.
Dec–Feb	Snowiest months (about 9 in of water)	Push salt-free plowed snow into the driveway-edge basins, never onto young plants; note where drifts form; brush heavy snow off young limbs; let roof snow slide into basins.
Mar–Apr	Snowmelt plus spring rain (about 4.8 in): the main harvest	Walk the block during melt and after rain-on-snow; log where water runs and ponds; check that basins empty within about a day; repair breached berms; stay off soft rows; open the gutters to the tanks.
May–Jun	Last dependable rain (about 3 in); bloom and leaf-out	Start the drip only when the soil 6–12 in down is drying (check with a trowel or probe); widen crescents; top up chips; mow and blow the alleys onto the rows.
Jul–Sep	Nearly dry (about 1.2 in in all); demand peaks (ET_0 about 7.9 in in July)	Drip runs inside the basins, topped up from the roof tanks while they last; after any thunderstorm, check for erosion and where the runoff went.
Late Sep–Oct	Reassess (principle 8)	Widen basins toward the drip line, fix trouble spots, drain tanks and hoses before hard freezes, and choose the next project (spreader, driveway basins, lane, corner basin).

Sources and Assumptions

- Brad Lancaster, *Rainwater Harvesting for Drylands and Beyond*, Vol. 1, 3rd ed. (Rainsource Press, 2019): eight principles (Ch. 1), site water budget (Ch. 2), earthworks and cistern overview (Ch. 3), "Raised Paths, Sunken Basins" (Ch. 4), calculations (App. 3), windbreaks and snow-drift harvesting (App. 8).
- Brad Lancaster, *Rainwater Harvesting for Drylands and Beyond*, Vol. 2, 2nd ed. (2019): infiltration basins (Ch. 2), berm 'n basins including "Harvesting Windswept Snow" (Ch. 3), infiltration trenches (Ch. 5), mulching (Ch. 7), hardscape runoff (Ch. 8), diversion swales including "Snow Diversion Berms" (Ch. 9), greywater (Ch. 12); water, A-frame, and laser levels (App. 1).
- Brad Lancaster, Water Harvesting Principles (wording of the eight principles).
- PRISM Climate Group, Oregon State University: 1991–2020 monthly normals, 800 m grid, sampled at Prairie (43.50° N, 115.57° W): about 22 in a year.
- Open-Meteo Historical Weather API (ERA5 reanalysis), 1995–2024 at the same point: FAO-56 reference evapotranspiration, and about 52% of precipitation falling as snow.
- Idaho Department of Water Resources, Adjudication Memorandum #11, Private and Diffused Waters: a landowner may use diffused surface water from rain and melting snow before it enters a natural channel (even one that flows only in spring), unless intercepting it measurably injures a prior water right.
- UC IPM: Bacterial Canker (cherry), worse in low or sandy spots, after spring freezes, and with drought stress or irrigation that wets the tree; Site Selection (Fruit), pears tolerate wet soils better than other fruit trees, while cherries and peaches cannot tolerate saturated soil; Phytophthora Root and Crown Rot, avoid prolonged saturation or standing water at the base of trees.
- USDA NRCS Web Soil Survey report for the property (soil folder): map unit 149, Simonton loam, 2 to 12 percent slopes, Elmore County Area, Idaho (version 14, August 31, 2026); layer data (Ksat, available water, clay) from NRCS Soil Data Access.
- CoCoRaHS: volunteer network that measures daily rain and snow at home with a standard gauge.
- Terrain: USGS 3D Elevation Program (3DEP) 1 m lidar, fitted to the fence lines on USGS NAIP imagery (331 × 1,292 ft fence to fence) and smoothed to about 5 ft; contours, flow paths, and slopes come from that surface, the same data as the whole-site maps (data/site_terrain.js). Bumps smaller than about 5 ft don't show. An earlier version of this page traced the owner's USGS 1 m contour screenshots; across the planted block the two agree within a degree of fall direction and about 0.1 ft per 100 ft of grade. Both replace the earlier field estimate (about 25 ft of fall to the north over 600 ft and 5–10 ft to the west): the north fall in the strip is closer to 10 ft, and the west tilt is about as strong.
- Block placement (owner's tape, October 24, 2026): R1 trunks on the south line of the 600 ft orchard strip, 67 ft north of the middle line, so R15 is about 320 ft from the north fence. The October 6 placement had R1 15 ft inside that line.
- Plan geometry (pollination grid snapshot, October 4, 2026): offset grid of 18 ft by 20 ft, C1 trunks 38 ft from the west fence, 35 ft windbreak (mulberry 5 ft and hazelnut 20 ft from the fence), alley 25–32 ft, orchard plot 35–272 ft, driveway 300–322 ft, east fence 332 ft with the elderberry hedge 5 ft inside it. The planted block is taken as 237 × 300 ft.
- Rough factors: a mature semi-dwarf tree uses about 0.8 × ET_0 over its 360 sq ft and roots 3–4 ft deep; the soil tank uses the Simonton layer values, about 6.4–8.1 in of available water in that depth. The owner reports no rock and about 10 ft of soil to rock.

Snowmelt and Storm Observation Log

Fill this in during snowmelt (March–April), after rain-on-snow, and after summer thunderstorms. Lancaster's first and last principles in practice: observe, then adjust. Use row–column positions (for example R7-C4) or landmarks.

Date	Event	Where water ran or ponded	Gone within a day?	Snow or frost notes	Fix or follow-up

Prairie Plants · Orchard rain and snowmelt harvesting plan · created October 6, 2026 · companion to orchard_pollination_grid.html, which this page does not modify.